
Federated Solar-Flare Prediction on SWAN-SF: A Provenance, Leakage, and Evaluation-Protocol Audit

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Abstract — We audit federated learning on the public SWAN-SF solar-flare benchmark. A provenance audit against the raw benchmark (normalisation-invariant instance matching, 98.7–100% coverage, 100% test-side label agreement) shows that the cleaned export’s same-partition train/test pairing shares instances: every flaring test window (6,234 of 6,234) also appears in training, and 85.7–90% of training positives are synthetic. On a leakage-free, temporally-preceding fold, discrimination generalises for every arm (ROC-AUC 0.906–0.978) and the federated-versus-centralised gap disappears; deployability is decided instead by persistence baselines and calibration — no arm beats same-region label inertia (TSS 0.967), only XGBoost (0.848) and logistic regression (0.489) clear 24-hour-lagged persistence (0.418), and no validation-fit decision layer survives the $26\times$ prevalence shift. Two apparent federated failures turn out to be BatchNorm artefacts: the in-partition FedAvg collapse (0.575) is a composite evaluation artefact, and on the raw, unbalanced substrate removing BatchNorm alone restores the federated MLP (FedAvg 0.963/0.366, FedProx 0.975/0.345, the latter matching the centralised MLP on ROC-AUC within recorded retrain nondeterminism), while pooled-statistics recalibration — validated as an instrument on the centralised model — collapses every federated arm to chance, a real finding. This is the first federated evaluation of SWAN-SF and the first instance-level provenance audit of its cleaned export. Every number regenerates from released result artefacts.

Keywords: federated learning; solar flare prediction; SWAN-SF; benchmark audit; data provenance; evaluation protocol; BatchNorm; class imbalance

Source code and all experimental artefacts: <https://github.com/Daveshvats/Federated-Learning-for-Distributed-Space-Weather-Monitoring>

1 Introduction

Solar-flare prediction from photospheric magnetogram time series is a benchmark-driven machine-learning problem. The community’s public reference is SWAN-SF [1], and the literature on it has grown around a small number of evaluation conventions: randomised splits of pooled windows, a cleaned and rebalanced export, and a reporting culture built on ROC/PR summary statistics. Those conventions carry risk. The benchmark’s own creators warned that sliding-window temporal coherence “invalidates the randomness assumption, thus impacting all sampling practices” [2]; subsequent work has echoed the warning [3], and the cleaned export’s release paper explicitly prescribes pairing temporally-preceding training partitions with a later test partition [4]. Flare-prediction models built on photospheric vector-magnetogram summaries — the SHARP parameter suite [5] — have demonstrated skilful classification of flaring versus non-flaring active regions [6, 7]; the predictive signal lives in quantities such as total unsigned current helicity, magnetic free energy, and flux-emergence proxies.

Why federate a public benchmark whose data is freely poolable? Methodologically, cross-silo federated learning [8, 9] is the standard instrument for studying non-IID data heterogeneity under a controlled protocol: it prescribes exactly what moves between data holders (model parameters) and exactly what does not (raw observations), which makes it a clean lens for asking how much of a benchmark’s apparent difficulty survives when the data is deliberately fragmented. Operationally, the SHARP-style parameters at the centre of this pipeline are dominated by a single public instrument (SDO/HMI via JSOC); the six-regional-agency narrative sometimes attached to federated space-weather proposals does not describe this data and is not claimed here. The six clients of this study are a simulation protocol — Dirichlet label skew across disjoint shards — not a claim about any institution. Federated flare forecasting itself is not new (a federated transfer-learning preprint exists [10]); what has not existed is a federated treatment of SWAN-SF that audits its own benchmark artefacts before interpreting its numbers.

This paper provides that treatment, and its character is an audit. We train six simulated clients on the benchmark in both its cleaned/rebalanced and raw, naturally-imbalanced forms, under a frozen evaluation protocol whose every selection decision is made on validation data, and we ask three questions: what does a federated pipeline actually achieve on SWAN-SF, what do the benchmark’s own artefacts do to the answer, and which parts of the observed behaviour belong to the algorithms rather than to the evaluation machinery? The contributions are:

1. **An instance-level provenance audit of the cleaned SWAN-SF export.** Normalisation-invariant matching against the raw benchmark (98.7–100% coverage, 100% test-side label agreement) shows the cleaned export’s same-partition train/test pairing shares instances: every flaring test window (6,234 of 6,234) is in the training pool, and 85.7–90% of training positives are synthetic (Section 3). The leakage phenomenon is documented prior art [2, 4]; the instance-level quantification inside a federated study is new.
2. **A leakage-free federated evaluation.** On the temporally-preceding fold (train partitions 1–4, single-pass test on partition 5), all arms generalise (ROC-AUC 0.906–0.978) and the in-partition federated-versus-centralised gap disappears; event-level detection is essentially solved for every arm, and false-alarm burden — not detection — separates them (Section 5).
3. **Corrected persistence baselines as the deployability verdict.** Same-region label inertia reaches TSS 0.967 with no learned model; only XGBoost (0.848) and logistic regression (0.489) clear 24-hour-lagged persistence (0.418) on TSS, and no arm clears it on HSS (Section 5).
4. **Two BatchNorm findings that re-attribute apparent federated failures.** The in-partition FedAvg collapse (0.575) is a composite evaluation artefact — untransported normalisation statistics compounded by a saturated validation monitor and the checkpoint rule it feeds; and on the raw, unbalanced substrate, an architecture-only control (three BatchNorm layers replaced by the identity, nothing else changed) restores the federated MLP to the centralised model’s ROC-AUC level, so the collapse is dominated by BatchNorm under federation, not by the flattened-feature encoder and not by federation per se (Section 5).
5. **A calibration failure that is real, not an artefact.** Pooled-statistics recalibration — validated as an instrument on the centralised model, where it reproduces the model’s own statistics to within 7×10^{-5} ROC-AUC — collapses every federated arm to chance on the raw substrate, so a pooled-statistics transport fix would not rescue federation there; and no validation-fit decision layer survives the $26\times$ prevalence shift between the balanced training partitions and the natural-prevalence test (Section 6).

6. **A frozen, reproducible protocol.** Immutable splits, validation-only selection and calibration, single-pass test evaluation, and machine-readable artefacts for every number reported here, released with the code.

We make no operational alerting claim. The benchmark’s data is public and freely poolable; the downstream flare→CME→storm→GIC chain is physically far from the quantity modelled and is not evaluated (see [11] for that literature).

2 Related work

Three literatures meet in this study. First, flare prediction from magnetogram summaries: the SHARP parameter suite [5] established the feature vocabulary; benchmark comparisons [6, 7] established the verification culture (TSS, HSS, reliability, persistence baselines) that Section 5 applies; and the FLARECAST programme [7] established the operational framing. SWAN-SF itself [1] contributed the partitioned benchmark with multivariate time-series windows, and its cleaned release [4] contributed the rebalanced export whose provenance Section 3 audits. The benchmark’s creators have been explicit about the sampling hazard since 2019 [2], and the cleaned release prescribes temporally-preceding folds — warnings this paper quantifies at the instance level and then obeys.

Second, federated learning. FedAvg [8] and FedProx [12] are the workhorse algorithms evaluated here; the cross-silo survey of Kairouz et al. [9] frames the methodology. The known interaction between federation and normalisation layers is the FedBN line of work [13]: Batch-Norm running statistics do not transport with parameters, and the interaction can masquerade as a federated optimisation failure. Section 5 shows two instances of exactly that masquerade on this benchmark, one of them compounded by a saturated drift monitor. Class imbalance under federation connects to focal loss [14] and its federated adaptations [15], and to oversampling approaches [16] whose value on this benchmark the extended record quantifies as a clean negative at natural prevalence.

Third, federated space weather. Federated flare forecasting exists as a 2023 transfer-learning preprint [10]; the present study is, to the authors’ knowledge, the first federated evaluation of SWAN-SF, the first instance-level provenance audit of its cleaned export, and the first leakage-free re-evaluation of federated flare prediction under the benchmark’s intended protocol. Federated learning has proven transformative in settings where privacy constraints really are binding [17]; no such constraint is claimed for this public data, and the federation here is a controlled heterogeneity instrument rather than a privacy solution.

3 The benchmark, its cleaned export, and the leakage audit

SWAN-SF and the two substrates. SWAN-SF [1] partitions solar-active-region time series into five folds of 60-minute, 60×24 multivariate windows over 24 SHARP-style magnetic-flux parameters, with a binary M/X-class flare label on every window. The community mostly consumes its *cleaned* export [4]: under-sampled negatives, Tomek-cleaned boundaries, TimeGAN-augmented positives, imputed values, and per-partition normalisation — a pre-balanced substrate (48.9% training prevalence) whose natural-prevalence test partitions carry 1.31–1.88% positives. This study evaluates on both substrates: the cleaned export as shipped, and a raw substrate rebuilt in-pipeline from the raw benchmark (the release’s imputation and normalisation reproduced from their published description, every parameter fitted on training data only, verified against the released export on the audit-matched windows, series differences bounded by 5×10^{-3} at float16 storage). The raw substrate removes the synthetic positives and the $26 \times$

prevalence shift and asks the classical question: can the models learn the natural distribution at all?

The provenance audit. The cleaned export carries no region or event identifiers, so the audit first aligns every cleaned window to its raw instance. The raw benchmark (Harvard Dataverse) encodes, per instance, the flare class, the HARP region identifier, and window timestamps in its filenames. The matching key is the per-feature argmax/argmin position along the 60 timesteps — 48 discrete invariants that survive any per-feature monotone normalisation and are robust to the export’s imputation noise. A window is accepted as aligned when at least 28 of 48 invariants agree (borderline cases additionally require mean per-feature Spearman $\rho \geq 0.70$). On the test partitions this verifies 98.7–100.0% of windows with 100.00% label agreement; on the training side, label agreement is 99.71% (161 disagreements, which bound the synthetic-share estimate below).

What the audit found. Three structural facts follow. First, the test export of partition p contains *all* raw instances of p ; the counts match exactly on every partition. Second, the training export of p is a rebalanced *subset of the same instances*: the verified training pool covers 56,005 unique raw instances, every one of which is also a test instance (56,005/56,005) — and every M/X-flaring test instance (6,234 of 6,234, 100% of the minority class) is in the training pool. Third, 85.7–90.0% of the training positive windows are TimeGAN-synthetic (they have no raw counterpart), while virtually no negative windows are. The release paper prescribes pairing temporally-preceding training partitions with a later test partition; a pipeline that instead pools the training exports and evaluates on the same partitions’ test exports — the default drift of any naive consumer — partially measures memorised data, most severely for the minority class. Because no HARP region spans two partitions (verified: 0 shared HARPs), a temporally-preceding fold is automatically instance-, region-, and time-disjoint.

The leakage-free fold. All headline comparisons are therefore re-run under the benchmark’s intended protocol: training on the training exports of partitions 1–4 (77,865 windows; 65,406 train / 12,459 validation after a stratified carve), and a single-pass evaluation on the test export of partition 5 (75,365 windows, natural 1.31% prevalence). The raw substrate uses the same partition structure, so its fold is leakage-free by the same construction. Everything else is frozen: six Dirichlet-sketched clients, 50 rounds, seed 42, the focal loss and calibration machinery of Section 4. Figure 1 shows the client partition the federated arms actually train on.

4 Federated protocol

Simulated cross-silo federation. Six simulated clients (neutral labels A–F; no institution participates) hold disjoint shards of the training pool, partitioned by a Dirichlet($\alpha=1.0$) label-skew draw. Each round, a random subset of clients trains the shared architecture locally; the server averages the returned *parameters* (size-weighted FedAvg [8]); FedProx [12] adds the proximal term ($\mu = 0.01$) to the local objective. Raw observations never move — the transport contract that makes the heterogeneity study controlled. The local objective is a server-aware federated focal loss [14, 15] with the focusing parameter clamped for stability under extreme imbalance.

Architectures. The MLP arm consumes the standard flattened feature encoding of the 60×24 window (six summary statistics per magnetic parameter; 144 features) through a three-layer network with BatchNorm and dropout (29,377 parameters). The architecture control of

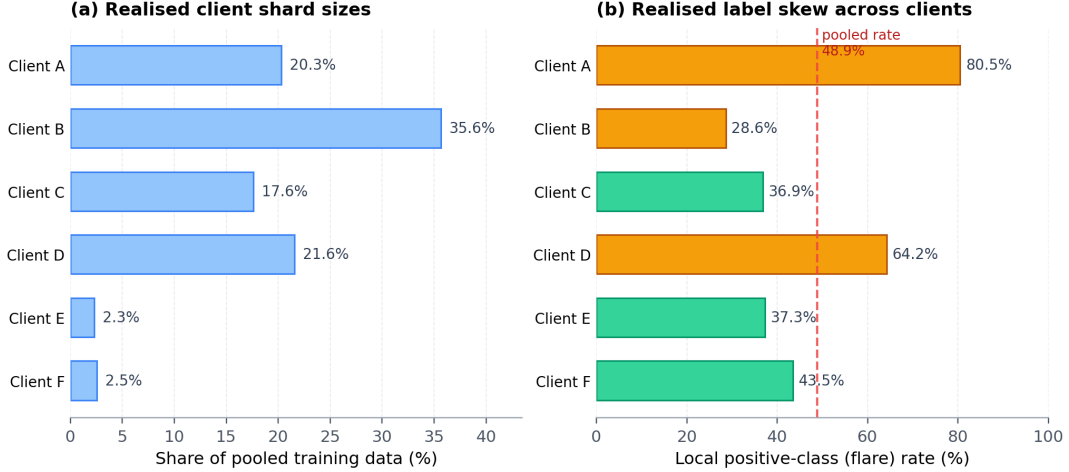


Figure 1: Non-IID partitioning of the pooled training data into six simulated clients by label-aware Dirichlet sampling (concentration $\alpha = 1.0$, seed 42). Proportions are read directly from the frozen run’s cached client assignment: the shard sizes and label skews are the heterogeneity every federated arm of this paper trains under.

Section 5 replaces the three `BatchNorm1d` layers with the identity and changes nothing else (28,929 parameters) — no running statistics exist, so parameter transport is exact. A sequence arm (SolarLSTM, 219,265 parameters, consuming the raw 60×24 series directly, no BatchNorm) provides the BatchNorm-free architecture comparator; its full study lives in the extended record. Centralised references (logistic regression, XGBoost, the pooled MLP and LSTM of identical architecture) are trained on the pooled data under the same budget conventions.

Frozen evaluation contract. Splits are immutable. Threshold and calibration selection happen on validation only; test evaluation is single-pass. Operating points are reported two ways: at the validation-selected F_2 -optimal threshold, and at validation-frozen fixed-FPR targets (0.5/1/2/5%), which are invariant to monotone miscalibration and to prevalence. Neural posteriors pass a prior-shift calibrator fit on validation before thresholding. Checkpoint selection is the best-validation- F_1 rule, with the selected round and a validation-ROC-selected sensitivity row disclosed beside it and per-round test trajectories recorded as diagnostics; no test-based selection happens anywhere. Verification metrics (TSS, HSS, Brier skill, 10-bin reliability, climatology) accompany every stored operating point, and persistence baselines are computed on the true time order within each active region, as the verification culture mandates.

Reproducibility. Every number in this paper regenerates from machine-readable result artefacts committed with the code; the underlying dataset cache is frozen under a 20-file SHA-256 manifest verified by a released script, and an independent re-execution from the public artefacts reproduces the in-partition headline metrics exactly. The extended record in the repository carries the full apparatus: five-seed statistics, the component-ablation grid, client-level holdouts, the sequence-arm and oversampling studies, and the audit trail of every protocol defect found and fixed along the way.

Table 1: Leakage-free fold (train partitions 1–4, single-pass test on partition 5, natural 1.31% prevalence) against the shipped in-partition numbers. Threshold-conditional calibration metrics use the validation-frozen prior-shift pipeline; ROC/PR are threshold-free.

Model	In-partition		Leakage-free fold				
	ROC	PR	ROC	PR	Brier	ECE	R@2%FPR
Logistic regression	0.824	0.235	0.978	0.455	0.375	0.477	0.695
XGBoost (centralised)	0.977	0.493	0.971	0.457	0.055	0.101	0.692
Centralised MLP	0.888	0.153	0.973	0.382	0.544	0.687	0.555
FedAvg MLP	0.575	0.199	0.906	0.370	0.127	0.337	0.652
FedProx MLP	0.954	0.307	0.976	0.308	0.271	0.508	0.631

5 Results

5.1 The leakage-free fold: discrimination survives, the federation gap does not

Table 1 compares the five headline arms on the leakage-free fold against the shipped in-partition numbers. Three observations follow. First, *discrimination generalises*: every arm attains at least 0.906 ROC-AUC on the unseen partition, and the task is easier than the in-partition protocol suggested (logistic regression reaches 0.978/0.455). Second, the centralised-versus-federated gap of the in-partition protocol *disappears*: the pooled MLP of identical architecture reaches 0.973/0.382 against FedProx’s 0.976/0.308 — a 0.003 ROC difference and a reversed PR ordering. The in-partition gap (0.888 vs 0.954) was an artefact of the protocol pairing, not a federation effect. Third, the FedProx-versus-FedAvg contrast only partially survives: FedProx leads by 0.070 ROC-AUC but trails FedAvg by 0.062 PR-AUC, and the next subsection shows why neither difference should be read as an algorithm property while BatchNorm sits in the loop.

The calibration and threshold-transfer failures persist leakage-free — the balanced-validation carve (48.9% prevalence) again faces the natural-prevalence test (1.31%), the same 26-fold prior shift without the instance overlap — so they are genuine properties of the balanced-training/natural-test interface rather than overlap artefacts: the validation-frozen FPR operating points are violated by every arm (XGBoost realises 15.3% FPR at the frozen 2% target; FedProx 49.4%), and only XGBoost’s posteriors remain usable (Brier 0.055 versus 0.127–0.544 for the neural arms).

Persistence baselines set the deployability floor. The toolkit of Leka et al. [6] — TSS, HSS, persistence, climatology — accompanies every operating point, and the persistence baselines set two floors that are not decorative. Same-region previous-window persistence (label inertia) achieves TSS 0.967 / HSS 0.970 on the partition-5 test with no learned model at all: at a 24-hour label horizon on ~60-minute windows, an active region that is flaring is still flaring, and *no arm in this paper — XGBoost included — beats that floor*. The 24-hour-lagged variant (forecast = the label of the same-region window nearest $t - 24$ h) scores TSS 0.418 / HSS 0.434; on TSS, XGBoost (0.848) and logistic regression (0.489) are the only arms above it, and every neural arm at its frozen threshold sits below it, while on HSS no arm clears the floor at all (XGBoost’s HSS is 0.201). This is the classic Leka-era result, and a benchmark audit is the right vehicle for stating it plainly: at this window geometry the honest question is not whether models beat persistence, but whether anything beats *lagged* persistence — and on this fold, at TSS, two arms do, none at HSS. A committed sensitivity sweep over eight candidate lag definitions agrees: the seven point-lag variants land at TSS 0.392–0.418 with the same two arms

Table 2: Event-level metrics on the leakage-free fold (65 true M/X-class events, validation-frozen F_2 thresholds, 1-hour alert cooldown). FA = false alarm.

Model	Events detected	FA windows/day	Alerts/day	Lead to peak (h)
Logistic regression	65/65 (100%)	8.6	8.8	23.4
XGBoost (centralised)	64/65 (98.5%)	1.7	1.9	23.3
Centralised MLP	65/65 (100%)	12.7	12.8	23.4
FedAvg MLP	65/65 (100%)	11.6	11.8	23.4
FedProx MLP	65/65 (100%)	11.9	12.1	23.4

Table 3: Raw-substrate retraining (train raw partitions 1–4, natural 2.05% prevalence, imputation and normalisation reproduced in-pipeline with train-only parameters; single-pass test on raw partition 5, natural 1.31% prevalence; identical frozen protocol). The leakage-free cleaned-substrate fold is reproduced for comparison.

Model	Fold (cleaned)		Raw substrate			
	ROC	PR	ROC	PR	Brier	R@2%FPR
Logistic regression	0.978	0.455	0.978	0.448	0.053	0.732
XGBoost (centralised)	0.971	0.457	0.974	0.329	0.018	0.587
Centralised MLP	0.973	0.382	0.971	0.445	0.010	0.596
FedAvg MLP	0.906	0.370	0.875	0.056	0.013	0.063
FedProx MLP	0.976	0.308	0.930	0.149	0.032	0.343

clearing each, while a lookback-flare variant scores 0.964 — label inertia under another name, which no arm clears.

Event-level evaluation. Regrouping the partition-5 test decisions into physical flare events (65 true M/X-class events, validation-frozen thresholds, 1-hour alert cooldown; Table 2), all five arms detect effectively every event (64–65 of 65) with a median first-alert lead of 23.4 h before flare peak. That lead must be read for what it is: the label is positive for every window in the 24 hours before a flare, so a median first alert near the horizon’s edge measures the label boundary, not precursor physics. What separates the arms is their false-alarm burden: XGBoost emits 1.7 false-alarm windows per day versus 11.6–12.7 for the neural arms — a $7\times$ difference that mirrors the window-level calibration gap and confirms that the deployability boundary on this benchmark is calibration, not detection.

5.2 The raw substrate: an apparent federated collapse

Retraining the identical frozen protocol on the raw, unbalanced benchmark separates the arms decisively (Table 3). The centralised arms are substrate-robust (logistic regression unchanged to the third decimal; the centralised MLP’s PR-AUC *rises* from 0.382 to 0.445; XGBoost’s falls from 0.457 to 0.329 — the synthetic positives were inflating its minority-class precision, not teaching it). The federated MLP arms are not: FedAvg falls to 0.875/0.056 with its calibrated posteriors never crossing the validation-frozen threshold (test recall 0.000), FedProx to 0.930/0.149, and FedAvg detects *zero* of the 65 events. Read naively, this is a federation failure. The next two subsections show it is mostly a BatchNorm artefact.

5.3 BatchNorm finding 1: the in-partition collapse is a composite evaluation artefact

The client MLPs contain three `BatchNorm1d` layers, and local training updates their running statistics every batch — but the federation transports *parameters only*: running statistics are

Table 4: Test ROC-AUC / PR-AUC of the identical in-partition FedAvg global weights under four BatchNorm treatments (frozen protocol, seed 42). Round 20 is the state the shipped checkpoint rule returned. Arm D is what a buffer-transporting implementation would have evaluated.

BN treatment	Round 5	Round 20	Round 50
A: init buffers (shipped)	0.950 / 0.264	0.575 / 0.199	0.098 / 0.010
B: recalibrated (pooled)	0.786 / 0.044	0.893 / 0.099	0.881 / 0.086
C: batch statistics	0.946 / 0.328	0.951 / 0.329	0.953 / 0.288
D: transported round buffers	0.885 / 0.157	0.931 / 0.194	0.935 / 0.209

buffers, not parameters, and buffers never move. The server-side global model therefore evaluates with its initialisation buffers forever (the shipped checkpoints carry `num_batches_tracked` = 0, `running_mean` = 0, `running_var` = 1 after 50 rounds), where each BN layer degenerates to the fixed affine map $x\gamma + \beta$. The second half of the mechanism is the saturated monitor: with init buffers, evaluated probabilities squeeze into a narrow band, validation F_1 pins to its all-positive closed form $2p/(1+p)$, and the best-validation checkpoint rule — acting on that saturated signal — restored the round-20 FedAvg state over the healthy round-5 state.

Table 4 decomposes the published in-partition FedAvg numbers: the frozen 50-round protocol was re-run faithfully and the identical global weights evaluated at three rounds under four BN treatments. The verdict is unambiguous. The weights never lost discriminative content: under evaluation-time batch statistics the round-50 state — whose shipped evaluation is 0.098, below chance — holds 0.953 ROC-AUC, and the round-20 state the rule shipped holds 0.951. The in-partition collapse to 0.575 is a composite evaluation artefact: untransported normalisation statistics — a mechanism documented since FedBN [13] and analysed in subsequent BatchNorm-under-federation studies [18–20] — compounded by a saturated monitor and the checkpoint rule it feeds. A controlled no-BN re-training in-partition confirms the training-side reading: with the three BN layers removed and nothing else changed, neither FedAvg nor FedProx diverges, the validation monitor stays informative, and the 0.379 ROC-AUC gap between the two shipped artefacts shrinks to 0.002–0.036 at every round.

5.4 BatchNorm finding 2: removing BatchNorm restores the federated MLP on the raw substrate

The decisive control applies the same intervention to the raw substrate: retrain both federated MLP arms with the three `BatchNorm1d` layers replaced by the identity, changing nothing else (same frozen protocol, shards, budget, aggregation, seed). Table 5 reports the outcome. FedAvg gains +0.088 ROC-AUC and +0.310 PR-AUC (6.5×); FedProx gains +0.045 and +0.196. In gap terms: FedAvg’s ROC gap to the centralised MLP closes from 0.096 to 0.008 (91%) and its PR gap from 0.389 to 0.079 (80%); FedProx’s ROC gap closes *past* zero (−0.004 residual) and its PR gap from 0.296 to 0.100 (66%). Both no-BN arms sit level with their BatchNorm-free sequence counterparts on ROC-AUC (FedAvg 0.963 vs the sequence arm’s 0.958; FedProx 0.975 vs 0.970), and no-BN FedProx *matches* the centralised MLP on ROC-AUC (0.975 vs 0.971): the −0.004 residual sits inside the retrain nondeterminism the replication record documents for exactly these arms (gate deltas −0.002 to +0.023 ROC-AUC), so the single-seed reading is parity within noise, not federated precedence. The raw-substrate collapse of the BN-carrying MLP arms is therefore dominated by BatchNorm under federation — not by the flattened-feature encoder, and not by federation per se. Two boundaries keep the claim honest: (a) PR-AUC, not ROC-AUC, is where work remains (no-BN FedProx’s 0.345 sits below the centralised MLP’s 0.445 and the sequence arm’s 0.404) — what these arms license is the attribution, not a new state-of-the-art row; (b) the level claims now carry a second seed: re-executed at seed 43 (initialisation and shard draw reseeded, validation carve unchanged), the shipped selections move by at most 0.002 ROC-AUC

Table 5: No-BatchNorm control on the raw substrate (single intervention: three BatchNorm1d layers $\rightarrow$ identity, nothing else changed; reference rows read from the released artefacts).

Arm (raw substrate)	ROC-AUC	PR-AUC
FedAvg MLP (BatchNorm, published)	0.875	0.056
FedAvg MLP (no BatchNorm, this control)	0.963	0.366
FedProx MLP (BatchNorm, published)	0.930	0.149
FedProx MLP (no BatchNorm, this control)	0.975	0.345
Centralised MLP (reference)	0.971	0.445
FedAvg LSTM (reference, no BatchNorm)	0.958	0.305
FedProx LSTM (reference, no BatchNorm)	0.970	0.404

(FedAvg 0.963 $\rightarrow$ 0.961, FedProx 0.975 $\rightarrow$ 0.976) and at most 0.024 PR-AUC without overturning an ordering, while the thresholded validation- F_1 monitor is the seed-sensitive layer (best values 0.53 $\rightarrow$ 0.66 and 0.09 $\rightarrow$ 0.30) — at 2% prevalence the rank metrics, not the monitor, carry the conclusions (Section 7).

The validation divergence, honestly. FedAvg’s no-BN test ROC-AUC *declines* from 0.974 (round 5) to 0.963 (round 50) while its validation ROC-AUC and F_1 still improve (0.974 to 0.977; 0.345 to 0.527), so the shipped best-validation rule selects the weakest test-ROC round of its own monitored trajectory. A prevalence difference cannot explain this divergence: ROC-AUC is a rank statistic, invariant to class prevalence, so the validation curve’s 2.05% against the test fold’s 1.31% moves thresholded quantities, not the ranking. Two candidate mechanisms remain, both testable with the released region-disjoint split generator: the random validation curve shares active regions with the training shards (the same leakage class the provenance audit documents benchmark-wide), and partition 5 is the temporally latest fold, so late-round specialisation to partitions 1–4 can register as validation gain and test loss at once. The within-fold region-disjoint validation re-run of both arms is prepared in the released code and is the first remaining compute item; FedProx’s trajectory, for contrast, climbs monotonically from 0.931 to 0.975 — the proximal term’s stabilisation becoming visible precisely once BatchNorm is removed.

6 Calibration: the deployability boundary

The prior shift is first-class. The validation prevalence (48.9%) and the test prevalence (1.9–1.31%) differ by a factor of 26 because the cleaned benchmark ships pre-balanced training partitions and natural-prevalence test partitions. The frozen protocol forbids using test labels — including test prevalence — for any selection, so the prior-shift calibrator is fit with the validation prevalence as its deployment target. The consequence is quantitative: at the frozen thresholds, FedProx’s calibrated Brier (0.264) and ECE (0.496) are *worse than the climatology’s* (0.239, 0.470), i.e. at natural prevalence the neural posterior is not deployment-usable as a probability, while the XGBoost reference suffers the least (Brier 0.063, ECE 0.115). Rebalanced training data invalidates naive threshold transfer: an F_2 -optimal threshold at 49% prevalence is semantically meaningless at 1.9%. The correct deployment instrument, pending full recalibration, is the recall-at-fixed-FPR operating point, which is invariant to monotone miscalibration and to prevalence.

The calibration-arm comparison is a clean negative. Five calibration arms (raw, prior-shift, Platt, isotonic, temperature), each fit on the balanced validation set and applied once to the test set, with the arm selected by minimum validation Brier — a proper scoring rule, so the selection itself is honest. The outcome: minimum validation Brier selects isotonic for all three neural models (validation Brier 0.012–0.025), and the selection is *wrong* — the selected

arms’ test Brier (0.352–0.541) is the worst of the five, because the balanced validation set rewards aggressive recalibration that the natural-prevalence test then punishes. No validation-fit decision layer survives the shift.

Pooled recalibration collapses federation: a real finding. Could a statistics-transport fix rescue the BatchNorm-carrying federated arms? The BN diagnostic’s pooled arm (B) — exact per-client statistics pooled by the law of total variance — is the oracle for exactly that fix, and on the raw substrate it *collapses* the federated arms to chance (0.500–0.501), even though in-partition the same oracle *recovered* FedAvg from 0.575 to 0.893. The instrument itself is validated: applied to the centralised raw MLP, where it should reconstruct the model’s own statistics, it reproduces the ROC-AUC to within 6.8×10^{-5} (0.96864 vs 0.96870). So the collapse is a real finding, not a bug: pooling exact per-client normalisation statistics across six heterogeneous raw-prevalence clients inflates variance past the point where any single pooled normalisation ranks the test distribution. A pooled-statistics transport fix would not rescue federation on the raw substrate — removing the BN layers (or transporting per-client buffers) is the class of fix that works, and the no-BN control of Section 5 is its direct proof.

What survives, end to end. Ranking survives everywhere for the centralised arms and, once BatchNorm is removed, for the federated arms. Probabilities survive only for XGBoost. Thresholds frozen at balanced prevalence transfer for nothing. Same-region label inertia remains unbeaten at TSS. The deployability boundary is calibration, and on this benchmark it is drawn by the prevalence shift the cleaned export itself introduced.

7 Discussion and limitations

What this audit changes. For benchmark consumers, the instance-level audit turns a documented warning [2, 4] into a measured quantity: same-partition pairing on the cleaned export puts 100% of the minority test class into training, so in-partition numbers on that pairing partially measure memorisation. The leakage-free fold — the benchmark’s own intended protocol — is the honest instrument, and on it the apparent federated-versus-centralised gap vanishes. For federated-learning practitioners, the two BatchNorm findings are the lesson: normalisation layers whose statistics do not transport can manufacture both a catastrophic collapse and a stabilisation narrative (FedProx’s apparent safety was, in large part, compatibility with the untransported evaluation path), and a saturated validation monitor converts an evaluation defect into a checkpoint-selection defect. The engineering conclusion is BatchNorm plus buffer transport, or BatchNorm-free architectures — not BN removal as a general remedy, since the BN architecture is genuinely useful under correct normalisation.

Limitations and remaining work. The honest caveats, each with the instrument that would upgrade the claim: the leakage-free fold is a single fold and seed (fold replication breadth is prepared in the released runners); the no-BN control’s level claims — sequence-arm parity and centralised-MLP parity — sit inside the recorded retrain noise band, a boundary its second-seed replication confirms (both arms within 0.002 ROC-AUC of their seed-42 selections); the random validation carve inside the raw-substrate protocol may share active regions with the training shards, and the region-disjoint re-run that separates that hypothesis from temporal drift is likewise prepared in the released code (roughly 1.7 kiloseconds of GPU compute for both arms); the event-level results are single-fold with Wilson half-widths of ± 2.8 –4.0 detection points; the calibration analysis shows no validation-fit decision layer survives the prior shift, but prior correction toward an *estimated* deployment prevalence from unlabelled data remains unstudied; and the SCAFFOLD implementation measured in the extended record departs from

the reference algorithm in disclosed ways (a sign-inverted control-variate update among them), so its cells there are implementation-case evidence and no conclusion of this paper rests on them. Cryptographic privacy, poisoning defence, and the downstream GIC chain are out of scope.

Conclusion. Federated solar-flare prediction on SWAN-SF, evaluated honestly, is neither a failure nor a win: it is a mirror for the evaluation protocol. On the leakage-free fold every arm generalises and federation costs nothing measurable; on the raw substrate the apparent federated collapse is BatchNorm’s, not federation’s, and removing the layers restores parity with the centralised model; the real failures are inherited from the benchmark — a prevalence shift that no validation-fit decision layer survives, and a persistence floor that no learned model beats at this label geometry. The audit instruments themselves — instance-level provenance matching, the BN-transport diagnostic, the instrument-validated pooled-recalibration oracle — are released and reusable, and they are the contribution most likely to outlive this benchmark.

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